

Q. In Tableau, how can clustering be useful for business decision-making?

Ans:

Clustering in Tableau is used to automatically group similar data points based on their characteristics. It helps businesses identify patterns, customer segments, trends, and anomalies without manually defining groups.

Example

Suppose a retail company has customer data containing:

- Age
- Income
- Annual Spending

Using Tableau's **Cluster** feature, customers can be grouped into segments such as:

- High Income, High Spending
- High Income, Low Spending
- Low Income, High Spending
- Low Income, Low Spending

How Clustering Helps in Business Decision-Making

1. Customer Segmentation

Businesses can identify different customer groups and create targeted marketing campaigns.

Example: Offer premium products to high-spending customers.

2. Identify High-Value Customers

Clustering helps find customers who generate the most revenue.

Example: A bank can identify customers likely to purchase investment products.

3. Detect Outliers and Anomalies

Clusters make unusual data points easy to spot.

Example: Detect suspicious insurance claims that differ significantly from normal claim patterns.

4. Improve Resource Allocation

Organizations can focus resources on the most profitable or high-risk groups.

Example: Hospitals can identify departments with similar patient volumes and allocate staff accordingly.

5. Better Forecasting and Strategy

Understanding patterns within clusters helps businesses make informed strategic decisions.

Example: An e-commerce company can tailor promotions based on purchasing behavior clusters.